

Question	Answer	Marks	Guidance
5(a)	charge / potential (difference)	M1	Ratio must be clear. Allow 'p.d.' or 'voltage' for 'potential difference'. Ignore any symbols unless they are defined. Ignore any reference to units.
	charge is charge on one plate and potential is p.d. between the plates	A1	Allow 'across <u>both</u> plates' or 'across <u>parallel</u> plates' for 'between the plates'. For the A1 mark, must be in terms of p.d. or potential difference, not just voltage.
5(b)	$E = V_X = V_Y$	B1	Must be an explicit statement. Allow in words provided all three terms in the expression are clear. Allow V or V_T for E Allow in terms of V_1 and V_2
	$Q_T = Q_X + Q_Y$	M1	Must be an explicit statement. Allow in words provided all three terms in the expression are clear. Allow Q for Q_T Allow in terms of Q_1 and Q_2
	$C_T V = C_X V + C_Y V$ <u>leading to</u> $C_T = C_X + C_Y$ or $Q_T / V = Q_X / V + Q_Y / V$ <u>leading to</u> $C_T = C_X + C_Y$	A1	Allow any one of E , V_T , V_X , V_Y for V provided same symbol is used for all three terms. Allow the first part left in terms of E , V_X and V_Y if the explicit B1 mark statement is clear. Everything required for the A1 mark must be in terms of C_T , C_X and C_Y ,

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5(c)	$E = \frac{1}{2} CV^2$	C1	<u>Independent C1 marks</u> – neither implies the other, and XP in one does not negate the other. Allow $E = \frac{1}{2} QV$ and $Q = CV$
	$C_T = (1/C + 1/2C)^{-1}$ or $1/C_T = 1/C + 1/2C$ $(C_T = \frac{2}{3} C)$	C1	C1 is for the working leading to $\frac{2}{3} C$. $\frac{2}{3} C$ does not need to be seen, but its appearance implies C1.
	$1.30 \times 10^{-3} = \frac{1}{2} \times \frac{2}{3} C \times 6.00^2$ $C = 1.08 \times 10^{-4} \text{ F}$ $= 108 \mu\text{F}$	A1	This line of working implies both C1 marks. Correct value of C in F implies both C1 marks. Correct to at least 3 significant figures. (AFC). Must be in μF for the A1 mark.